

Re-analysis Summary Report for Peer-evaluation

Paper ID: Adida_CompPolitStu_2016_G0Kb

Re-analyst's ID: 25fm1

Your task

As a peer-evaluator, for this paper you are asked to judge whether the analyst's applied analytical choices for Task 1 and Task 2 are acceptable. By acceptable, we mean that the analysis pipeline is within the variations that could be considered appropriate by the scientific community in addressing the given analytical task. In addition, for Task 1, we ask you to judge whether the reported conclusion adequately follows from the results of the analysis; whether the co-analyst's self-categorization of the result is adequate. Although, you are not required, it would be appreciated, if you could execute the accompanying analysis code to check any mismatch between the results of the analysis and the reported results.

For both tasks, you should provide your evaluation via this survey: <https://forms.gle/EAREaC6JjN2PWFfR6>

The re-analyst's task

The re-analyst was instructed to assess the following claim of the original authors:

... priming the first lady's ethnicity increases support for President Yayi among her coethnics. (p. 635.)

The corresponding article

Here you can find the original article: https://osf.io/hcxdq/files/osfstorage/5e21e2a6675e0e009c6b4d01/?view_only=c874fcf097644e82a320679d4641163a

and the original data: https://osf.io/avj8y/?view_only=978a4c4cd219427c9f529a1128478dc7

They are provided only as context of the re-analysis.

In Task 1, the analyst was asked to conduct the analysis without any restrictions. Here, the analyst reported the following steps and results of the analysis:

Since the codebook was missing, I first had to figure out to find which variables were to be taken into account in the analysis.

/! The Independent Variable "passage" (= experimental manipulation) includes 5 modalities in the dataset vs. 3 modalities reported in the paper. This discrepancy is of importance. Since there is no justification to remove two of the 5 conditions, they should be included in the data analysis. I contacted the author on May 4th to try to have more information about this, but I received no answer. In my opinion, removing experimental conditions without strong and sound justification is bad research practice. /!

I decided to conduct both analyses, with and without (as in the paper) the two conditions. Please see uploaded script for both analyses.

The DV : "BoniVote" (= intention of voting for President Yayi) is binary, we will use a logistic regression.

The IV : The first IV (experimental manipulation) has 5 modalities (passage: Control, Femme, Nago, Bariba, FemmeFon). The second IV (co-ethnicity) has 2 modalities (FonGroup: Yes, No).

A sum contrast (0.5 ; -0.5) was created for the IV FonGroup.

The dataset shows 5 modalities for the experimental scenario (passage: Control, Femme, Nago, Bariba, FemmeFon). Therefore, the design is a 5 (passage: Control, Femme, Nago, Bariba, FemmeFon) X 2 (FonGroup - coethnicity with first lady: Yes, No) between-subjects. The claim is an interaction : priming Fon ethnicity increases support for Yayi only among her co-ethnics. The predictions calls for the use of Helmert contrasts for the experimental condition. Control, Femme, Nago, Bariba, FemmeFon C1: -1 1 0 0 0 C2: -1 -1 2 0 0 C3: -1 -1 -1 3 0 C4: -1 -1 -1 -1 4 To be able to support the claim, we should observe an interaction between the contrast C4 and FonGroup.

This is the model to test the claim, written out with R syntax : `glm(BoniVote ~ passage*FonGroup, data=dt1, family = binomial(link = 'logit'))`

Here is the output :

```
Coefficients: Estimate Std. Error z value Pr(>|z|)
(Intercept) -0.18250 0.08380 -2.178 0.0294 *
passage1 -0.25734 0.14483 -1.777 0.0756 .
passage2 0.08448 0.07470 1.131 0.2581
passage3 0.01870 0.05322 0.351 0.7253
passage4 0.08737 0.03939 2.218 0.0265 *
FonGroup1 -1.92382 0.16760 -11.479 <2e-16 ***
passage1:FonGroup1 -0.01862 0.28967 -0.064 0.9487
passage2:FonGroup1 0.28882 0.14940 1.933 0.0532 .
passage3:FonGroup1 0.03505 0.10643 0.329 0.7420
passage4:FonGroup1 0.13806 0.07878 1.753 0.0797 .
— Signif. codes: 0 ‘ ’ 0.001 ‘ ’ 0.01 ‘ ’ 0.05 ‘ ’ 0.1 ‘ ’ 1
```

We observe a main effect of FonGroup, suggesting that being of the ethnicity Fon decreased voting intention for President Yayi, $z = -11.48$, $p < .001$ ($< .05$).

We also observe a main effect of the 4th Helmert contrast, suggesting that the experimental condition increased voting intention (vs. the four other conditions), $z = 2.22$, $p = .027$ ($< .05$).

However, the interaction between the 4th contrast (passage4) and co-ethnicity is not significant at the 0.05 significance level, $z = 1.753$, $p = .080$ ($> .05$). This interaction was necessary to be significant in order to support the claim " ... priming the first lady's ethnicity increases support for President Yayi among her coethnics." Given the main effect of "passage4" and the absence of significant interaction between "passage4" and "FonGroup", it appears the priming First Lady's ethnicity increasing voting intention overall. However, the data does not allow to claim that it is because of coethnicity with the First Lady. Therefore, my conclusion differs from the claim put forward by the authors of the paper.

The analyst's conclusion: There are high discrepancies between description of the manipulation in the paper and the dataset. Removing experimental conditions without justification is considered as a questionable research practice in my field (experimental social psychology). I chose to include all available modalities in the analysis. It appears the priming First Lady's ethnicity increasing voting intention overall. However, the data does not allow to claim that it is because of coethnicity with First Lady. The claim "priming the first lady's ethnicity increases support for President Yayi among her coethnics" is not supported by my analysis. Instead, it can be said that "priming the first lady's ethnicity increases support for President Yayi".

The analyst's categorization of the result:

The results do not show evidence for or against the relationship/effect as described in the claim provided in your task

In Task 2, the analysts received the following instruction to conduct the analysis:

You should restrict your analysis to Benin-related survey experiment data.

Here, the analyst reported the following steps and results of the analysis:

Participants with missing values on the DV were excluded. /! the Independent Variable includes 5 experi-

mental conditions in the dataset vs. 3 experimental conditions reported in the paper. This discrepancy is of importance. The corresponding author was contacted regarding this matter but did not answer. This is a probable case of QRP (see <https://replicationindex.com/2015/01/24/qrrps/> “Failing to disclose experimental conditions”). Since there is no justification to remove the two conditions, they should be included in the data analysis. This is the analysis I conducted. Independent Variables were contrast coded as follows : The DV is binary, which calls for the use of logistic regression. The dataset shows 5 modalities for the experimental scenario (passage: Control, Femme, Nago, Bariba, FemmeFon) Therefore, the design is a 5(passage: Control, Femme, Nago, Bariba, FemmeFon) X 2 (FonGroup - coethnicity with first lady) between-subjects The claim is an interaction : priming Fon ethnicity increases support for Yayi only among her co-ethnics. That is, we should observe an interaction between the fourth contrast (C4, see below) and FonGroup The predictions calls for the use of Helmert contrasts for the experimental condition. The IV passage (experimental condition) was contrast coded as follows : Control, Femme, Nago, Bariba, FemmeFon C1: -1 1 0 0 0 C2: -1 -1 2 0 0 C3: -1 -1 -1 3 0 C4: -1 -1 -1 -1 4

The IV FonGroup (being a co-ethnic of First Lady) was contrast coded -0.5 (no coethnicity) vs. 0.5 (coethnicity).

As in Task 2, the model included the main effects of the IVs (which were contrast coded), plus all their interactions.

The test of interest (interaction between C4 and FonGroup) was not significant, $b = 0.138$, $z = 1.753$, $p = .0797$ ($> .05$). As a consequence, the data do not allow to conclude that priming First lady’s ethnicity raised voting intention among her co-ethnics.

Analysis language/software/code (optional to check):

Task 1: R

Task 2: R

Analysis codes: <https://osf.io/zrb2v/>